Mohammad Sheriff Mehmood

Senior Data Scientist · NLP & Generative AI Engineer

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Senior Data Scientist with 5+ years delivering production ML and NLP systems in healthcare, finance, and retail. Deep expertise in Generative AI (RAG pipelines, LLMOps), end-to-end MLOps, and scalable predictive modelling. Track record of measurable business impact: $1M+ revenue uplift, 40% churn reduction, 60% faster model iteration. GCP Certified ML Engineer · LeetCode 400+ problems solved (Python).

TECHNICAL SKILLS

Languages & Generative AI	Python (primary), SQL, PyTorch, TensorFlow, Scikit-learn, LangChain, Amazon Bedrock, OpenAI API (GPT models), RAG, LLMOps, Fine-tuning, Prompt Engineering, Sentence-Transformers
MLOps & Cloud	MLflow, Docker, AWS Step Functions, AWS CDK (IaC), AWS (SageMaker, Lambda, ECS/Fargate, API Gateway), GCP, CI/CD, Model Versioning & Governance
Document AI & Computer Vision	Amazon Textract (Queries/Forms/Tables), Amazon Comprehend (Custom Classification), OpenCV, YOLO, OCR Pipelines, Document Forensics
ML & Statistical Modeling	XGBoost, Random Forest, Logistic Regression, UMAP, K-Means Clustering, Feature Engineering, A/B Testing, Hypothesis Testing
Databases & Vector Stores	Operational DBs: PostgreSQL, DynamoDB, MongoDB | Vector/Retrieval: FAISS, Pinecone, Qdrant
Tools & Engineering	FastAPI, AWS (S3, SQS, Secrets Manager, IAM/KMS/WAF), REST API Design, pytest, Git

PROFESSIONAL EXPERIENCE

ValueLabs · Hyderabad, India
Senior Data Scientist – AI/ML Engineer | Mar 2025 – Present

Embedded with Businessolver's Data Science team as a full-stack ML Engineer; designed and shipped an automated document analysis pipeline leveraging computer vision, VLMs, and AWS-native tooling to process healthcare documents at scale.

- **Document Analysis Pipeline:** Designed an end-to-end multi-modal pipeline on AWS (Comprehend, Textract, S3, ECS) to classify and extract data from 47,000+ healthcare documents, cutting manual claim validation time by **[NEEDS USER INPUT]**%.
- **Query-Based Document Understanding:** Built a custom QA system fusing Textract OCR visual features with textual embeddings across varied document layouts, applying tailored post-processing to produce unified QA outputs.
- Deployed inference on AWS Lambda for scalable processing — achieving **[VALIDATE METRIC]** 93% extraction precision at **[NEEDS USER INPUT]**% recall.
- **Annotation & Monitoring UI:** Built an interactive React UI with REST API for data annotation/validation, real-time model metric visualization, and data-driven threshold tuning — streamlining stakeholder review cycles.
- **Stacking Ensemble Model:** Trained and tuned a stacking ensemble (Random Forest, XGBoost, LightGBM) via cross-validation, achieving 0.78 F1 and 0.81 AUC; applied SHAP and Gini impurity for model explainability.
- **Hybrid Census Analytics Engine:** Re-architected census analytics with hybrid LLM–Pandas processing, reducing LLM calls from 30–40 to 1 and eliminating token-limit bottlenecks through schema mapping and deterministic analytics.

- **Centralized LLM & cloud infrastructure:** By building reusable OpenAI client factories and AWS Secrets Manager integration with singleton/cached clients, unified fallback/configuration, secure credential management, and centralized observability.
- **Production-Grade Hybrid RAG Architecture:** Designed a tenant-isolated hybrid RAG system using Qdrant and OpenAI embeddings with semantic + BM25 retrieval, query rewriting, bounded agentic search, source diversification, and citation-grounded responses, serving **[NEEDS USER INPUT]** tenants at **[NEEDS USER INPUT]**s p95 latency.
- **RAG Evaluation & Hallucination Mitigation:** Built an evaluation harness for the RAG system tracking **[NEEDS USER INPUT]** (e.g. retrieval precision/recall, faithfulness, answer relevance) against a **[NEEDS USER INPUT]**-query gold set, reducing **[NEEDS USER INPUT]**.
- **RAG Caching & Cost Optimization:** Implemented **[NEEDS USER INPUT]** (e.g. semantic/response caching) for the RAG pipeline, reducing redundant LLM calls by **[NEEDS USER INPUT]**% and cutting average cost-per-query from **[NEEDS USER INPUT]** to **[NEEDS USER INPUT]**.

LTIMindtree · Pune, India
Senior Data Scientist – NLP Engineer | Jan 2024 – Mar 2025

- **GenAI Chatbot & RAG System:** Architected an internal RAG-based knowledge assistant (Gemini + FAISS Vector DB) serving 15+ team members, achieving **[VALIDATE METRIC]** 90% query accuracy (measured against a **[NEEDS USER INPUT]**-query eval set) and sub-2s response time for enterprise knowledge retrieval.
- **LLMOps Pipeline:** Built an end-to-end LLMOps pipeline using OpenAI, LangChain, ChromaDB, and Sentence-Transformers, cutting model iteration time by 60% (from **[NEEDS USER INPUT]** to **[NEEDS USER INPUT]**) and reducing deployment overhead by 50%.
- **Experimentation:** Ran A/B tests on **[NEEDS USER INPUT]** (e.g. retrieval strategy, threshold tuning), comparing **[NEEDS USER INPUT]** across control and treatment groups, informing the decision to **[NEEDS USER INPUT]**.

Cognizant Technology Solutions · Chennai, India
Programmer Analyst to Data Scientist | Dec 2019 – Jan 2024

Joined as Programmer Analyst core ML systems, promoted to Data Scientist (leading cross-functional AI/NLP initiatives.

- **ML Model Development:** Built and deployed supervised ML models, improving prediction accuracy by 20% and saving $200K in operational costs; improved baselines by 5–10% using Neural Networks, ensemble methods, GridSearchCV, and hyperparameter tuning.
- **AI Customer Support Platform:** Led a 7-member cross-functional team to build an NLP-powered support application, reducing customer churn by 40% **[VALIDATE METRIC]** (measured via **[NEEDS USER INPUT]**) and generating $300K in additional quarterly revenue through context-aware intelligent responses.
- **Fraud Detection System:** Reduced false-positive fraud alerts by 20% using Logistic Regression and Random Forest with class-weighted resampling; received the company Innovation Award.
- **Data Platform & BI:** Unified 10+ heterogeneous data sources—including databases, APIs, and third-party feeds—into ML-ready pipelines; built Tableau dashboards tracking 20+ KPIs, improving cross-team reporting efficiency by 30%.

RESEARCH PROJECTS

Accident & Fall Detection System
Python · YOLO · OpenCV · CNN · CUDA · SMTP

- Built real-time human fall detection system achieving 95% accuracy using YOLO + CUDA toolkit on edge hardware.
- Designed smart camera surveillance for highway accident detection; automated ambulance dispatch via SMTP alerts, cutting emergency response time by 40%.

Abstractive Text Summarisation

Python · BERT · NLP · Transformers

- Applied BERT and NLP toolkits to summarise live chat context, improving agent efficiency by 35% and reducing customer response times by 25%.
- Gave agents instant access to previous chat history — reducing handle time and improving resolution quality.

Visa Approval Forecasting (US Immigration)

Python · ML · MongoDB · Docker · AWS · GridSearchCV

- Predicted US visa approval outcomes using production-grade ML models; performed EDA, feature engineering, and hyperparameter tuning via GridSearchCV.
- Connected MongoDB for data persistence; containerised solution with Docker for AWS deployment.

Disease Classification Mobile App

TensorFlow · CNN · TFLite · FastAPI · GCP · ReactJS · ReactNative

- Achieved 92% accuracy detecting Early/Late Blight in potato crops; quantized CNN with TF Lite — reduced model size by 80% and inference latency by 50% for on-device deployment.
- Deployed serverless inference on Google Cloud Functions (25% cost reduction); served real-time predictions via TF Serving + FastAPI to ReactJS and React Native frontends.

End-to-End Topic Modelling & MLOps on AWS

Python · LDA · Docker · AWSECS · MLflow · CodePipeline · EC2

- Built LDA topic modelling pipeline on AWS ECS with blue/green deployment — cut processing time by 40% and deployment time by 60%.
- Automated full CI/CD (CodeCommit, CodeBuild, CodeDeploy, CodePipeline); reduced manual effort by 70%.
- Used MLflow for model versioning; ensured high availability via EC2 + ECS load balancing.

EDUCATION

M.Tech in Data Science & Engineering **BITS Pilani, India | Oct 2021 – Oct 2023**

B.E in Computer Science & Engineering **OIST, India | Aug 2015 – Jun 2019**

CERTIFICATIONS & AWARDS

1st Prize – Trizetto Hackathon (Patient Data Integration, 95% approval rate)	GCP Professional ML Engineer (Certified Feb 2023)
Certified MLOps Developer – Dataiku	IBM Professional Data Science Certification
Secretary, Toastmasters Club · Speaker, PyData Conference	OpenAI Whisper Hackathon – Hearing-impaired accessibility app